WEEK 7 - HEMATOLOGY • LECTURE 4 • SLIDE COMPANION

Approach to Anemia

Mark Dalgetty • designed to sit beside the 62-slide lecture deck

Official Week 7 LOs | Slide-order companion | ScholarRx links | Diagnostic algorithm emphasis | Current guideline notes separated | Factoid sheet companion

Overview: What the Official Week 7 Learning Objectives Want From You

LO 1. Describe a brief overview of RBC production.

LO 2. Define anemia and its etiologies.

LO 3. Describe defining signs and symptoms of various causes.

LO 4. Approach a patient with anemia to reach a diagnosis.

LO 5. Identify the different laboratory tests used for diagnosis.

LO 6. Identify available and cost-effective treatment modalities.

SCHOLARRX BRICKS PAIRED WITH THIS LECTURE

Anemias: Foundations and Frameworks
Anemias: Putting It All Together

Both are explicitly listed under Anemia in the official Week 7 objective sheet.

LO 1

Slides 3-9: ingredients for erythropoiesis, iron/hepcidin, EPO, RBC indices, reticulocytes, and oxygen delivery.

LO 2

Slides 10-17: practical definition plus the production/loss/destruction framework and MCV categories.

LO 3

Slides 12-14 + disease blocks: symptoms, physical signs, and cause-specific clues such as pica, jaundice, neurologic findings, and Burton line.

LO 4-5

Slides 12-20 + throughout: history first, CBC/differential, reticulocytes, MCV, smear, iron studies, hemolysis labs, kidney/liver testing.

LO 6

Slides 31, 35, 38, 39-49, 59-61: oral/IV iron, treat the cause, ESA logic, restrictive transfusion, B12/folate replacement.

SOURCE NOTE - IMPORTANT

The uploaded companion DOCX calls itself **Lecture 4: Anemia** but says it was based on a different lecturer and a 66-slide lecture. The actual uploaded deck is **Mark Dalgetty's 62-slide Approach to Anemia**. This guide uses Dalgetty's deck as the primary slide-order source and uses the DOCX only as supplemental explanation where it matches the current topic. I do not silently treat the two as identical.

THE SINGLE MOST IMPORTANT FRAMEWORK IN THIS LECTURE

Low hemoglobin -> CBC/differential -> reticulocyte response -> MCV -> targeted labs/smear.

The reticulocyte count asks **"Is the marrow responding?"** The MCV asks **"What size problem am I dealing with?"**

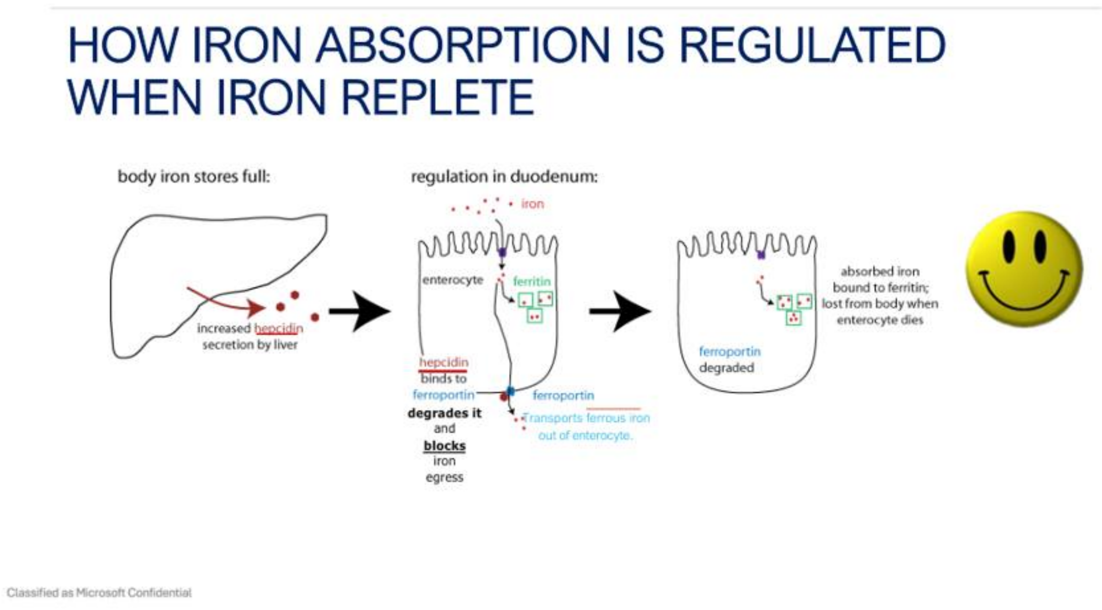
Slides 3-9

RBC production, iron handling, EPO, indices, and reticulocytes

LO 1 + LO 5

Slides 3-6: building an RBC requires substrate + signal + functioning marrow

- **Iron** is needed for heme/hemoglobin.
- **Vitamin B12 and folate** support DNA synthesis during erythroid precursor proliferation.
- **Erythropoietin (EPO)** is the kidney-derived signal that increases erythropoiesis when oxygen delivery is inadequate.
- The lecture uses iron regulation early because it becomes the mechanistic bridge to iron deficiency and anemia of inflammation later.



Lecture slide 5 - full slide retained. When iron stores are replete, liver-derived hepcidin binds ferroportin and drives its degradation, limiting iron export from enterocytes. This mechanism returns later in anemia of inflammation.

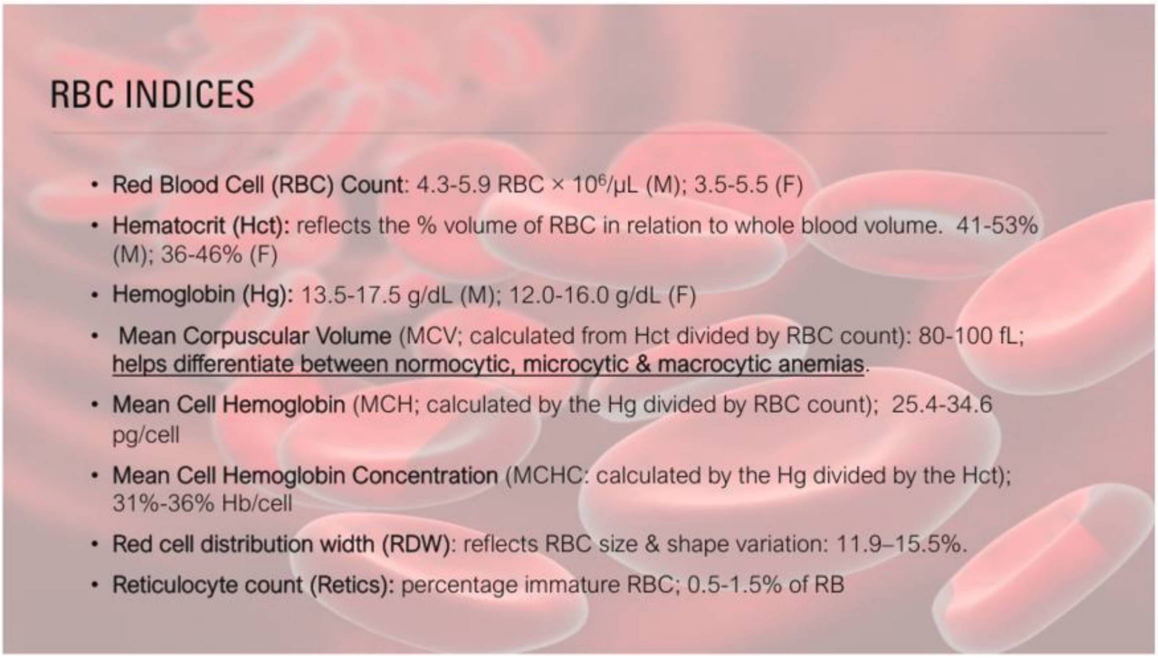
NEED TO MEMORIZE - HEPCIDIN

Hepcidin binds ferroportin -> ferroportin is degraded -> less iron reaches plasma.

High hepcidin = iron trapped in enterocytes/macrophages and reduced iron availability for erythropoiesis.

WHAT THIS MEANS CLINICALLY

Iron status is not just "how much iron exists in the body." It is also **whether iron can leave storage/absorptive cells and reach transferrin and marrow**. That distinction explains why anemia of inflammation can have low serum iron despite normal/high ferritin.

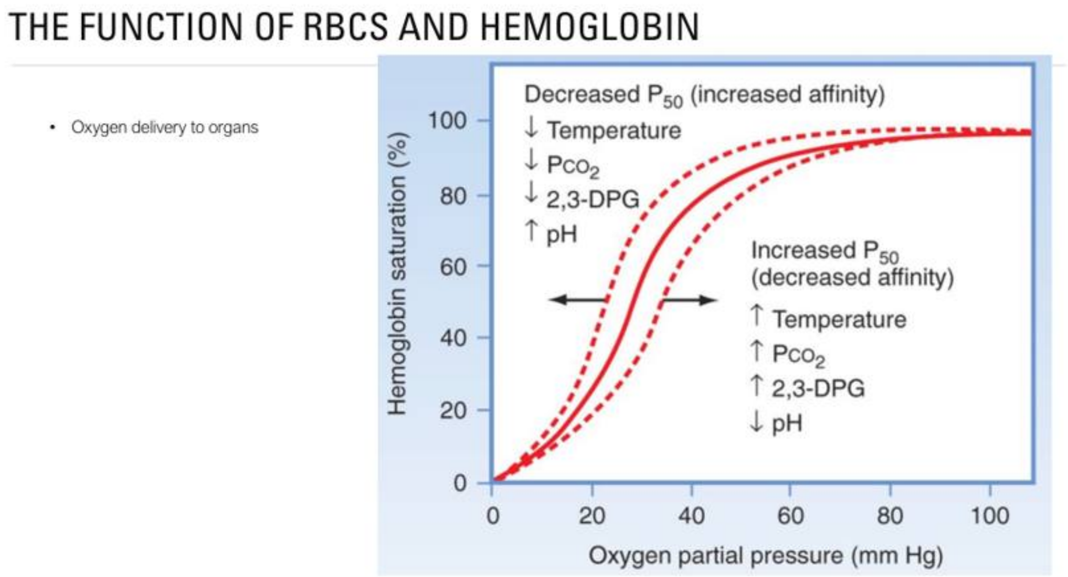


Lecture slide 7 - full slide retained. Dalgetty gives the numerical framework: Hb/Hct confirm anemia; MCV classifies cell size; RDW captures size variation; the reticulocyte count measures immature RBC output.

Index	Lecture value / meaning	How to use it
MCV	80-100 fL normal	<80 microcytic; 80-100 normocytic; >100 macrocytic.
MCHC	31-36%	Reflects Hb concentration per RBC; hypochromia lowers it.
RDW	11.9-15.5%	Higher = more size heterogeneity (anisocytosis); often rises in IDA.
Reticulocytes	~0.5-1.5% in the slide	Must be interpreted in context of anemia; a "normal" percentage can be inappropriately low when Hb is low.

FACULTY EMPHASIS - RETICULOCYTES

The deck explicitly warns that reticulocytes mature for about four days and can be released earlier during severe anemia. The point is diagnostic, not trivia: **reticulocytes tell you whether marrow output is appropriately increasing.**



Lecture slide 9 - full slide retained. Dalgetty uses the curve to explain compensation. A right shift (higher P50) helps unload oxygen; increased 2,3-DPG is one reason a slowly developing anemia can be tolerated better than an abrupt fall in Hb.

LO 1 checkpoint

Can you explain why iron, B12/folate, EPO, and marrow function are all required for erythropoiesis, and why reticulocytes are the best early clue to whether marrow is compensating?

Slides 10-20

Define anemia, take the history, and build the diagnostic fork

LO 2-5

Slides 10-11: practical definition, but context changes the meaning

- Dalgetty defines anemia as reduced oxygen-carrying capacity and practically uses **hemoglobin** rather than RBC count.
- Lecture thresholds: **female Hb <12 g/dL**; **male Hb <13 g/dL**.
- He immediately challenges universal cutoffs: pregnancy, age, altitude, and baseline physiology matter.

DON'T MISS THIS CLINICAL POINT

Symptoms depend on rate of decline, not just the final Hb. A rapid 12 -> 9 g/dL drop can be dramatic, while a patient with a slow chronic decline may compensate surprisingly well at a much lower Hb.

Slide 12: "History" appears five times for a reason

- Pica / ice craving, restless legs, heavy menstrual or GI bleeding -> iron deficiency.
- Recent medications -> consider oxidant hemolysis or other drug-related anemia.
- CKD, heart failure, chronic inflammatory disease -> underproduction / inflammation.
- Stomatitis/glossitis and diet/GI history -> nutritional deficiency.
- Family history -> inherited hemoglobin, membrane, or marrow disorders.

Slide 14: physical exam narrows mechanism

- **Pallor** = anemia clue, nonspecific.
- **Jaundice/dark urine** = think hemolysis.
- **Koilonychia / nail changes** = iron deficiency clue.
- **Thalassemia facies** = chronic marrow expansion clue.

Decreased RBC production	RBC loss	RBC destruction		
Iron deficiency	Menorrhagia	Antibody mediated	Congenital hemolytic anemias	Mechanical intravascular destruction
B12 deficiency	GI bleeding	AIHA	Thalassemia	DIC
Folate deficiency	Any other source of bleeding	Cold agglutinin disease	Sickle cell disease	Thrombotic microangiopathy
Sideroblastic anemia		Paroxysmal cold hemoglobinuria (PCH)	Glucose-6-phosphate dehydrogenase (G6PD) deficiency	Mechanical valves and intravascular devices.
Bone marrow toxicity e.g. alcoholism		Hemolytic transfusion reactions (acute and delayed)	Pyruvate kinase (PK) deficiency	
Myelodysplastic syndrome			Hereditary spherocytosis (HS)	
CHIP			Hereditary elliptocytosis (HE)	

Lecture slide 15 - full slide retained. The slide separates decreased production, RBC loss, and RBC destruction. This mechanism-based classification should sit above the MCV classification in your head.

MECHANISM FIRST

Production problem: marrow cannot make enough -> retic response is low/ inadequate.

Loss or destruction: marrow should respond -> retic response is high if marrow reserve is intact.

Slides 16-18: the lab sequence

1. **CBC + differential:** confirm anemia and look for other cytopenias/blasts.
2. **Reticulocyte count:** is marrow compensating?
3. **MCV:** microcytic vs normocytic vs macrocytic.
4. **Targeted labs:** chemistry/renal/liver, iron studies, hemolysis labs, nutritional tests.
5. **Peripheral smear:** morphology can collapse the differential quickly.

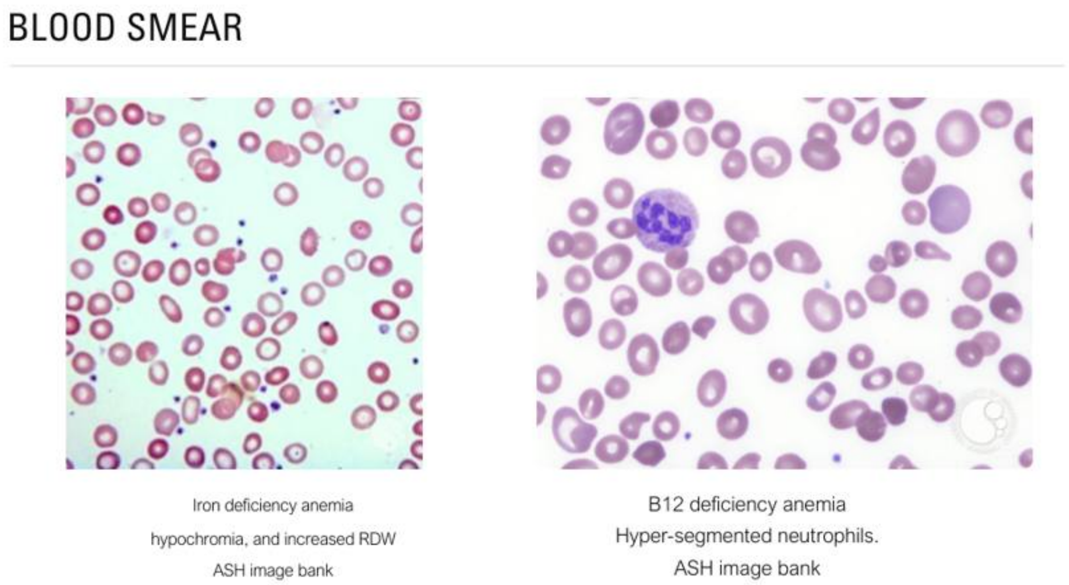
THEN CHECK MCV… IT CAN GIVE YOU SO MUCH INFORMATION ON THE POSSIBLE CAUSE.

Microcytic MCV < 80	Normocytic 80-100	Macrocytic > 100
Iron deficiency	Anemia of chronic disease	B12/Folate deficiency
Thalassemia	Anemia of CKD	Copper deficiency
Lead poisoning	Mixture of both causes of microcytic and macrocytic	MDS
Anemia of chronic disease	Blood loss anemia	Hemolytic anemia
		Hypothyroidism
		Any cause of increased retic count

Lecture slide 17 - full slide retained. Dalgetty's slide shows the overlap: anemia of inflammation can be microcytic or normocytic; blood loss is commonly normocytic; reticulocytosis itself can raise MCV because reticulocytes are larger.

BOARD ENRICHMENT - RETICULOCYTE PRODUCTION INDEX

In anemia, a raw reticulocyte percentage can look "normal" even when the marrow response is inadequate. Board-style questions often use a corrected reticulocyte count or reticulocyte production index. The conceptual rule is enough here: **appropriate high retic -> loss/destruction; inappropriately low retic -> production problem.**



Lecture slide 19 - full slide retained. This is a useful morphology contrast: iron deficiency shows hypochromic, variably sized cells; B12 deficiency shows macrocytes with hypersegmented neutrophils.

LO 4-5 checkpoint

A patient has low Hb. Can you state the first three decisions in order and explain what each one tells you: **CBC/differential -> reticulocyte response -> MCV**?

Slides 21-31

Iron deficiency anemia: symptoms, smear, iron studies, causes, treatment

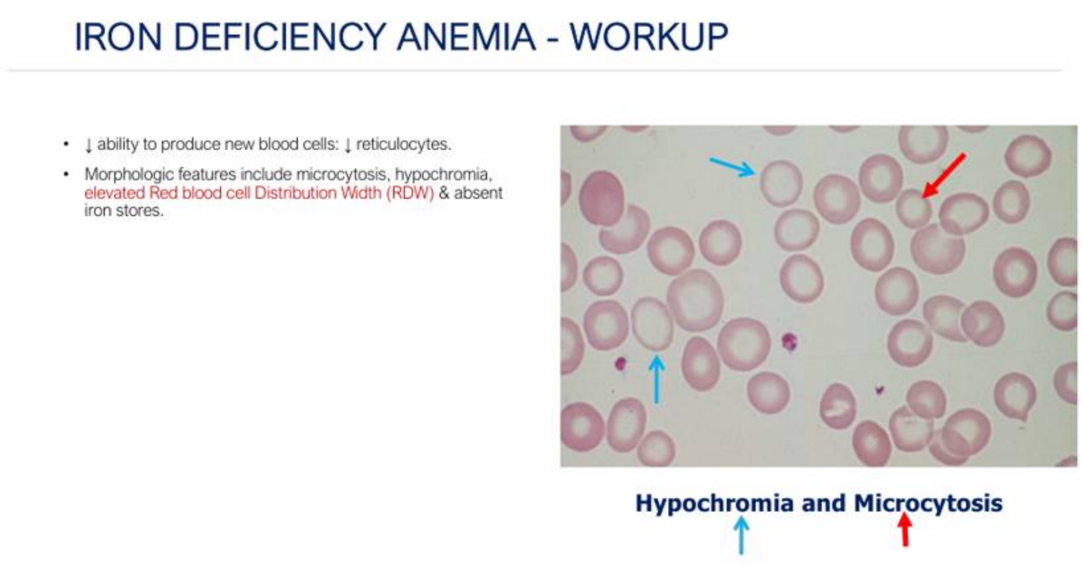
LO 2-6

Slides 21-25: high-yield recognition

- Iron deficiency is presented as the **most common anemia**.
- Iron is absorbed mainly in the **duodenum/proximal jejunum**; vitamin C improves absorption, while antacids/tannins/phytates can reduce it.
- Characteristic clues: **ice craving/pica, restless legs, fatigue, koilonychia**.
- Plummer-Vinson syndrome = **iron deficiency + dysphagia from a postcricoid esophageal web**.

PROFESSOR-STYLE CLINICAL REASONING

For IDA, diagnosing low iron is only half the job. **You must ask why the patient is iron deficient.** Menstrual blood loss, GI blood loss, pregnancy, donation, GU blood loss, malabsorption/bypass, and parasitic blood loss all appear in the deck.



Lecture slide 27 - full slide retained. The slide highlights low production, microcytosis, hypochromia, increased RDW, and depleted stores. The image is useful for central pallor and cell-size heterogeneity.

IDA lab	Direction	Why
Serum iron	Low	Less circulating iron bound to transferrin.
Ferritin	Low	Body iron stores are depleted.
TIBC / transferrin capacity	High	More unsaturated transferrin capacity is available.
Transferrin saturation	Low	Little of transferrin's capacity is occupied by iron.
RDW	Often high	Progressive deficiency produces greater size variation.
Retic	Low/inadequate before treatment	Substrate limitation prevents adequate RBC production.

NEED TO MEMORIZE - IRON STUDIES

IDA = iron down, ferritin down, TIBC up, saturation down.

Slide 31: treatment

- **Oral ferrous salt** is the low-cost first-line approach when tolerated/absorbed.
- The slide specifically calls ferrous sulfate inexpensive and widely available.
- Best absorption is on an empty stomach, but GI adverse effects are common.
- The lecture notes that hepcidin rises after a dose and presents **alternate-day dosing** as a strategy.
- Use IV iron when oral therapy is not tolerated, poorly absorbed, or clinically insufficient.
- **Treat the underlying source of iron loss.**

IDA checkpoint

Can you recognize IDA from pica/koilonychia + microcytic hypochromic cells, predict all four iron-study directions, and name the next clinical question after confirming it?

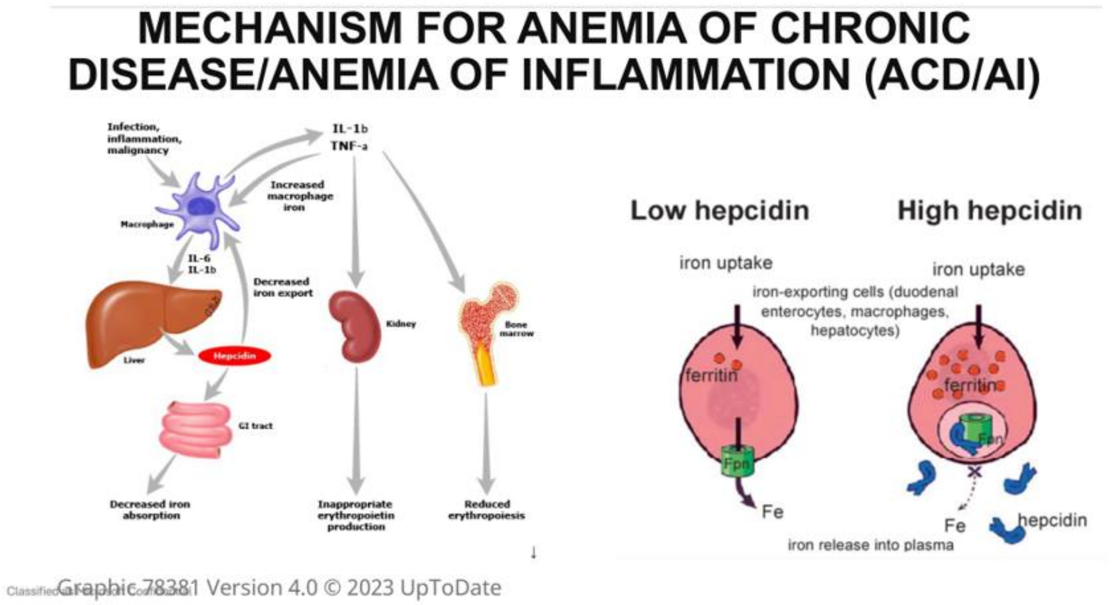
Slides 32-38

Anemia of inflammation and CKD: iron sequestration vs inadequate EPO

LO 2 + LO 4-6

Slides 32-35: anemia of chronic disease / inflammation

The lecture ties chronic infection, autoimmune disease, malignancy, obesity/inflammation, tissue injury, and chronic liver disease to a typically mild-moderate anemia.



Lecture slide 33 - full slide retained. This figure is mechanistically central: inflammatory cytokines increase hepatic hepcidin, which reduces intestinal iron absorption and macrophage iron release; erythropoiesis also becomes less responsive.

IDA VS ANEMIA OF INFLAMMATION

Both: serum iron down, saturation down.
IDA: ferritin down, TIBC up.
Inflammation: ferritin normal/high, TIBC down.

WHY FERRITIN CAN MISLEAD

Ferritin reflects storage iron but is also an **acute-phase reactant**. In inflammation, it can be normal/high even while serum iron available to marrow is low. This is functional iron restriction rather than simple depletion.

Slides 36-38: anemia of CKD

- Core mechanism: **reduced renal EPO output** -> inadequate erythroid stimulation.
- Dalgetty emphasizes checking retics, iron status, and CKD stage/eGFR and excluding other causes.
- The slide discusses iron replacement and ESA therapy when iron stores are adequate.

CURRENT 2026 GUIDELINE UPDATE - KDIGO

Dalgetty's treatment slide cites an older Hb target and a 2006 trial. The **KDIGO 2026 anemia-in-CKD guideline** now emphasizes individualized decisions about ESA or HIF-PHI initiation and warns against maintaining Hb above **11.5 g/dL** with ESA therapy because higher targets can increase harm. Treat this as current external enrichment, not as a replacement for what your professor may test from the slide.

KDIGO 2026 Anemia in CKD guideline hub

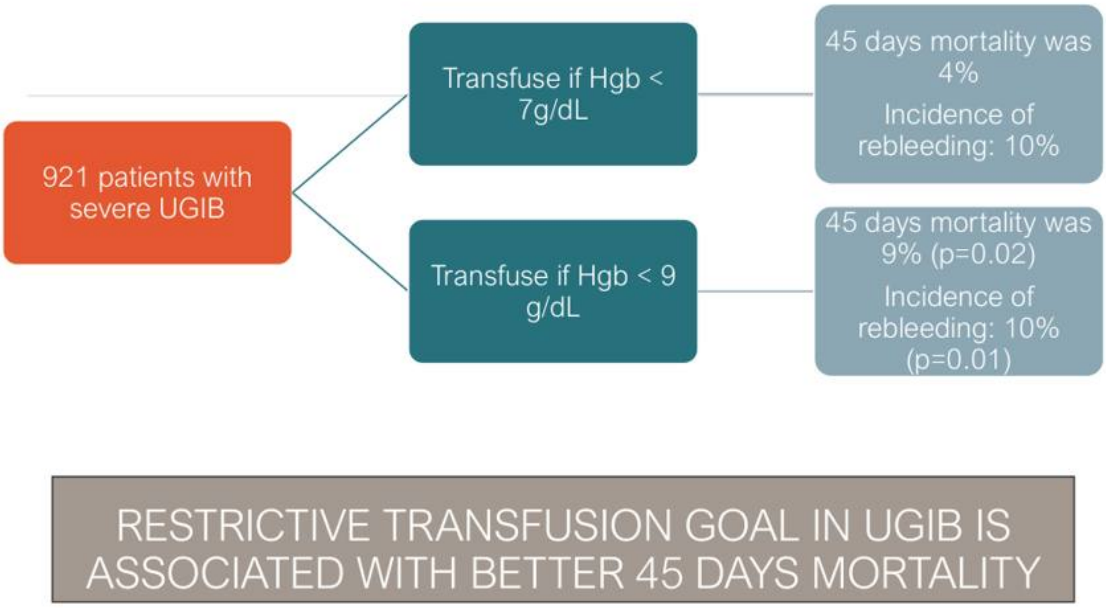
ACD/CKD checkpoint

Can you explain why inflammation causes **high hepcidin + low serum iron + normal/high ferritin**, and separately why CKD produces a low-retic anemia through inadequate EPO?

Slides 39-49 Acute blood loss and restrictive transfusion logic LO 3 + LO 6

Slides 39-42: acute blood loss is a physiology problem, not a number alone

- The deck gives a rule of thumb that ~500 mL blood loss can correspond to about a 1 g/dL Hb fall after equilibration, but the measured Hb changes with time and fluids.
- Clinical impact depends on **rate of loss, amount lost, IV fluid dilution, and marrow reserve**.
- In active bleeding, hemodynamics and ongoing blood loss matter more than waiting for a perfect Hb threshold.



Lecture slide 44 - full slide retained. Dalgetty highlights the trial comparing a restrictive 7 g/dL threshold with a 9 g/dL threshold in severe upper GI bleeding, with lower 45-day mortality in the restrictive group.

Slides 43-49: professor's transfusion take-home

- UGIB: slide summarizes a **7 g/dL** restrictive threshold.
- Septic ICU patients: slide summarizes **7 g/dL**.
- High cardiovascular-risk hip surgery: slide summarizes **8 g/dL** or symptomatic anemia.

CURRENT GUIDELINE CHECK - AABB 2023

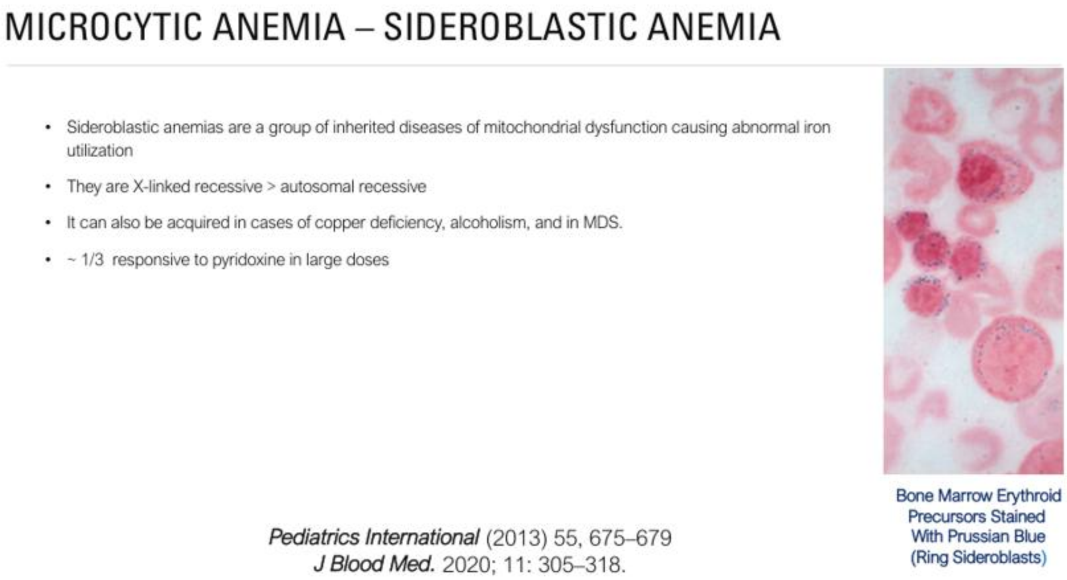
The lecture's restrictive framework remains aligned with modern guidance for **hemodynamically stable** adults. AABB 2023 recommends considering transfusion below **7 g/dL** for most stable hospitalized adults; clinicians may use **7.5 g/dL** for cardiac surgery and **8 g/dL** for orthopedic surgery or preexisting cardiovascular disease. Clinical context still matters.

2023 AABB International RBC Transfusion Guidelines

EXAM TRAP

A transfusion threshold is **not** a command to ignore shock, active hemorrhage, ischemic symptoms, or the patient's baseline risk. Dalgetty's case-based slides are teaching you to combine Hb with the clinical picture.

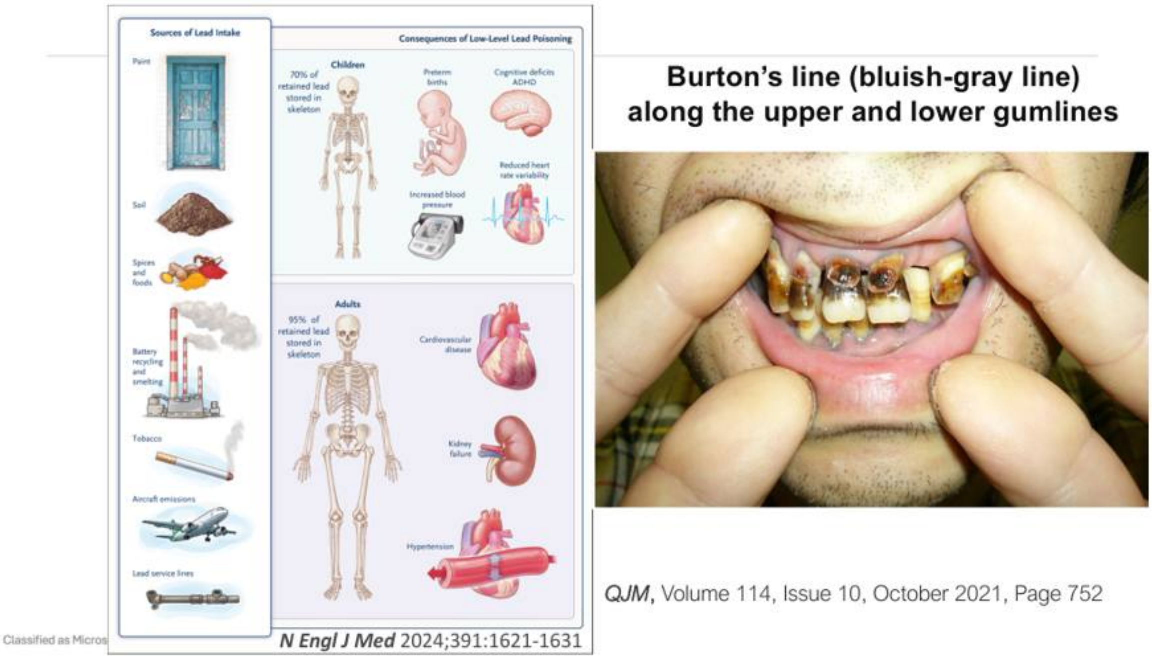
Slides 50-52 Sideroblastic anemia and lead poisoning LO 2 + LO 3 + LO 5



Lecture slide 50 - full slide retained. The key image is a Prussian-blue-stained marrow erythroid precursor with iron-laden mitochondria arranged around the nucleus: the ring sideroblast.

Slide 50: sideroblastic anemia

- Abnormal mitochondrial iron utilization / heme synthesis.
- Can be inherited or acquired; the slide lists **copper deficiency, alcohol, and MDS** among acquired settings.
- The slide notes that a subset responds to high-dose **pyridoxine (vitamin B6)**.
- Iron is present but cannot be appropriately incorporated -> **iron/ferritin tend to be high**.



Lecture slide 52 - full slide retained. The deck visually emphasizes environmental/occupational sources, neurologic/renal/cardiovascular effects, and the bluish-gray gingival Burton line.

LEAD - RECOGNITION SET

Abdominal colic + neuro/peripheral neuropathy + basophilic stippling + sideroblastic pattern.

The lecture also calls out **Burton line** at the gingival margin.

LINK BACK TO LECTURE 1

Lead inhibits **ALA dehydratase and ferrochelatase** in heme synthesis. That connection is deliberate Week 7 integration: the biochemical block from Heme Synthesis becomes a microcytic/sideroblastic anemia phenotype here.

Slides 53-61

Macrocytic anemia: megaloblastic vs non-megaloblastic, B12 vs folate, pernicious anemia

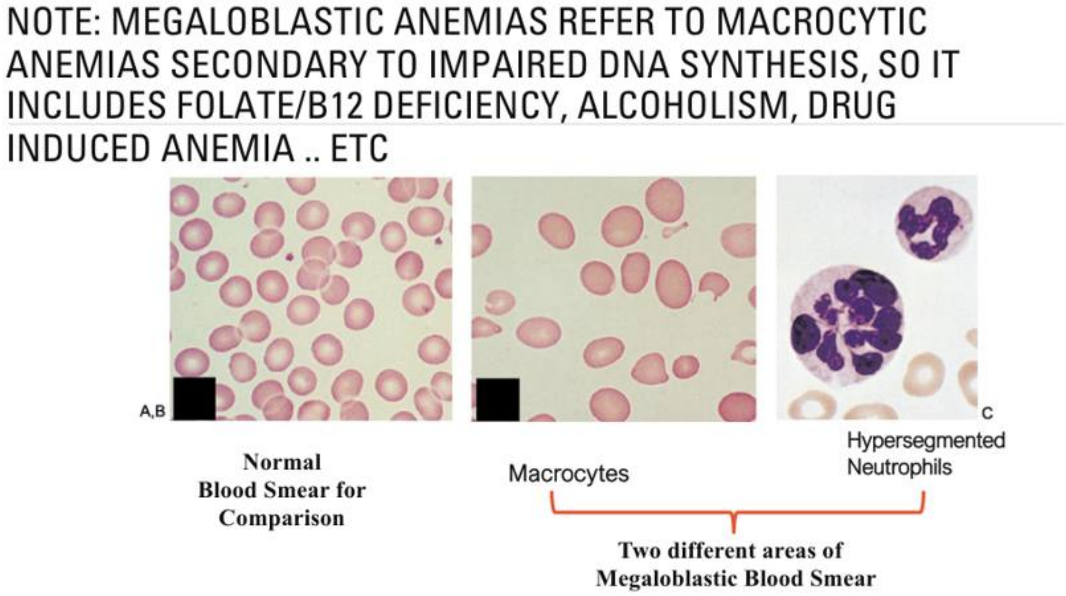
LO 2-6

Slide 53: macrocytosis is broader than B12/folate

- B12/folate deficiency.
- Reticulocytosis after RBC destruction (reticulocytes are larger).
- Alcohol.
- Drugs such as methotrexate, hydroxyurea, anti-epileptics.
- Liver disease.
- MDS.

FIRST SPLIT IN MACROCYTOSIS

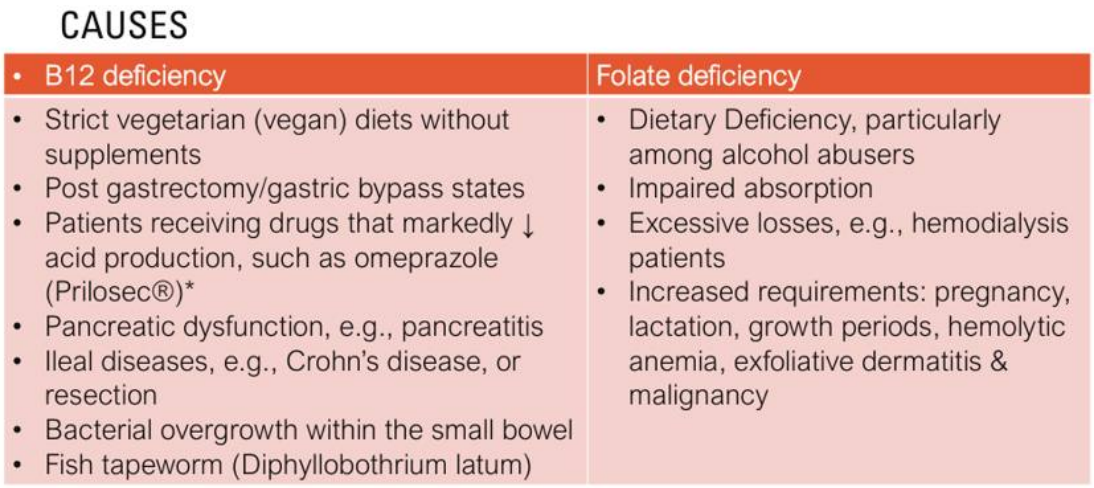
Megaloblastic = impaired DNA synthesis -> macro-ovalocytes + hypersegmented neutrophils.
Non-megaloblastic = other mechanisms such as alcohol/liver disease/ reticulocytosis, typically without the classic hypersegmented-neutrophil pattern.



Lecture slide 55 - full slide retained. Use the slide for the visual pair: macrocytes/macro-ovalocytes plus hypersegmented neutrophils. This is the morphology that should trigger B12/folate and DNA-synthesis questions.

Slides 54-58: B12 and folate overlap, but B12 owns the neurologic findings

- Both are needed for DNA synthesis and can cause ineffective hematopoiesis/ megaloblastic change.
- The lecture states B12 stores take **years** to deplete; folate stores take only **months**.
- **B12 deficiency:** paresthesias, loss of vibration/position sense, ataxia, spasticity, psychiatric/cognitive changes can occur.
- **Folate deficiency:** no characteristic subacute-combined-degeneration neurologic syndrome.



CAUSES

• B12 deficiency	Folate deficiency
• Strict vegetarian (vegan) diets without supplements • Post gastrectomy/gastric bypass states • Patients receiving drugs that markedly ↓ acid production, such as omeprazole (Prilosec®)* • Pancreatic dysfunction, e.g., pancreatitis • Ileal diseases, e.g., Crohn's disease, or resection • Bacterial overgrowth within the small bowel • Fish tapeworm (Diphyllobothrium latum)	• Dietary Deficiency, particularly among alcohol abusers • Impaired absorption • Excessive losses, e.g., hemodialysis patients • Increased requirements: pregnancy, lactation, growth periods, hemolytic anemia, exfoliative dermatitis & malignancy

Lecture slide 57 - full slide retained. The source slide separates B12 causes (vegan diet, gastric/ileal disease, bacterial overgrowth, fish tapeworm, etc.) from folate causes (diet/alcohol, malabsorption, losses, and increased demand).

Test / feature	B12 deficiency	Folate deficiency
MCV	High	High
Smear	Macrocytes + hypersegmented neutrophils	Same megaloblastic pattern
Homocysteine	High	High
Methylmalonic acid	**High**	Normal
Neurologic deficits	**Yes**	No classic B12-type neurologic syndrome
Stores	Years	Months

CLASSIC BOARD TRAP - DO NOT GIVE FOLATE ALONE IF B12 DEFICIENCY IS POSSIBLE

Folate can improve the hematologic abnormalities while **neurologic injury from untreated B12 deficiency continues**. The uploaded companion specifically emphasizes this masking phenomenon.

B12/FOLATE DEFICIENCY - TREATMENT

B12 deficiency	Folate deficiency
• Start intramuscular cyanocobalamin supplementation at 1mg weekly x 4 then switch to monthly. (Can start with daily injections in severe hospitalized patients) • Can also do oral supplementation with oral or sublingual Cyanocobalamin 1-2 mg daily	Start folate replacement at 1mg/day

Lecture slide 59 - full slide retained. Dalgetty gives practical replacement regimens: IM or high-dose oral/sublingual cyanocobalamin for B12, and oral folate replacement for folate deficiency.

Slides 60-61: pernicious anemia = autoimmune B12 malabsorption

- Anti-parietal-cell and/or anti-intrinsic-factor antibodies.
- Chronic immune-mediated gastritis with gastric atrophy.
- Association with other autoimmune diseases.
- Increased gastric carcinoma risk is explicitly listed.
- Very high-dose oral B12 can still work through passive absorption despite intrinsic-factor loss; parenteral therapy is also used.

Macrocytic checkpoint

Can you distinguish megaloblastic from non-megaloblastic macrocytosis and explain why **MMA + neurologic symptoms** distinguish B12 deficiency from folate deficiency?

LO 4-5 synthesis

The anemia algorithm you should actually use on a vignette

LO 4 + LO 5

STEP 0

Confirm low Hb and look at the rest of the CBC. Pancytopenia or blasts changes the urgency and differential immediately.

#	Question	If yes / high	If no / low
1	**Is the reticulocyte response appropriate?**	Think **loss or destruction**: bleeding, hemolysis.	Think **underproduction**: iron/B12/folate, CKD/EPO, inflammation, marrow disease.
2	**What is the MCV?**	Micro <80; normo 80-100; macro >100.	
3	**What does the smear show?**	Hypochromia/anisopoikilocytosis, hypersegmented neutrophils, target cells, bite cells, schistocytes, spherocytes, basophilic stippling all redirect the workup.	
4	**What targeted labs fit the mechanism?**	Iron/ferritin/TIBC/saturation; creatinine/eGFR; LDH/bilirubin/haptoglobin; B12/folate/MMA/homocysteine; inflammatory markers or marrow testing when indicated.	
5	**What caused it?**	Do not stop at "iron deficiency" or "B12 deficiency." Identify bleeding, diet, malabsorption, inflammation, kidney disease, drugs, marrow disease, inherited disease, etc.	

Microcytic

IDA: ferritin down/TIBC up.
ACD/AI: ferritin normal-high/TIBC down.
Thal: very low MCV with relatively preserved RBC count/normal iron.
Sideroblastic/lead: iron-loaded pattern + marrow/toxin clues.

Normocytic

High retic: bleeding/hemolysis.
Low retic: CKD/EPO failure, marrow suppression, inflammation, mixed disease.

Macrocytic

Megaloblastic: B12/folate/DNA-synthesis problem.
Non-megaloblastic: alcohol, liver disease, hypothyroidism, reticulocytosis, some marrow disorders.

Do not forget mixed disease

Two opposing processes can yield a "normal" MCV. RDW, smear, reticulocytes, and the clinical story can uncover mixed microcytic + macrocytic disease.

Learning-Objective Synthesis

Official LO	What you should be able to do without notes
LO 1	Explain iron/hepcidin/ferroportin, EPO signaling conceptually, and how indices/reticulocytes reflect RBC production.
LO 2	Define anemia practically and classify by production/loss/destruction and by MCV.
LO 3	Recognize rate-of-drop symptoms, IDA clues, hemolysis clues, B12 neurologic findings, and lead/sideroblastic clues.
LO 4	Use CBC -> retic -> MCV -> smear/targeted labs to build a diagnosis rather than ordering every test at once.
LO 5	Interpret iron studies, hemolysis labs, renal function, B12/folate/MMA/homocysteine, and smear findings.
LO 6	Choose low-cost oral iron when appropriate, treat the cause, use IV iron selectively, understand restrictive transfusion logic, and replace B12/folate appropriately.

One-table comparison

Condition	MCV / retic	Signature labs	Fast clue
Iron deficiency	Micro; retic low/ inadequate	Fe down, ferritin down, TIBC up, sat down	Pica/ice, koilonychia, bleeding source
Anemia of inflammation	Normo or micro; retic low	Fe down, ferritin normal/high, TIBC down	High hepcidin / chronic inflammatory setting
CKD	Usually normo; retic low	Reduced EPO physiology; iron status variable	Low eGFR + inadequate marrow response
Acute blood loss	Normo; retic rises after marrow response	Clinical bleeding/ hemodynamics	Rate and amount matter more than Hb alone
Sideroblastic	Often micro/ dimorphic	Iron/ferritin high; ring sideroblasts	Alcohol, MDS, copper, B6-responsive subset
B12 deficiency	Macro; low production	Homocysteine up, **MMA up**	Hypersegmented neutrophils + neuro findings
Folate deficiency	Macro; low production	Homocysteine up, MMA normal	Megaloblastic smear without B12-type neuro findings

LAST 60 SECONDS

Anemia workup = CBC -> retic -> MCV -> smear/targeted labs.
IDA: ferritin down, TIBC up. **Inflammation:** ferritin up, TIBC down.
CKD: EPO problem. **Blood loss/hemolysis:** retic should rise.
B12: MMA up + neurologic findings. **Folate:** MMA normal.
Restrictive transfusion: ~7 g/dL for many stable adults, but always integrate clinical context.

Source hierarchy: Dalgetty's 62-slide Anemia deck is the primary slide-order source. The uploaded Lecture 4 companion is supplemental and is not treated as identical because it names another lecturer and a 66-slide source. Official Week 7 objectives determine emphasis. Current AABB and KDIGO notes are clearly labeled external enrichment rather than blended into faculty content.